Tennessee Technological University

Department of Electrical and Computer Engineering

ECE 3050: Circuits II Laboratory

Experiment 6 Report

Author: Colby Alley

Lab Partner: Alek Presley

Lab Instructor: Ifeoluwa Adeloye

Date of Experiment: 3/25/2026

Date of Submission: 4/8/2026

I. Introduction

In Experiment 6, the application of diodes in half-wave rectification is studied. Rectification is the process where an AC signal is transformed into a signal with unidirectional current. While rectification can be categorized as half-wave or full-wave, this experiment specifically focuses on the half-wave configuration [1], [2], [3]. The objective of Experiment 6 is to utilize experimental data to verify LTspice software simulation, ensuring accuracy to real-world circuit behavior [2]. Since rectification plays a major role in linear power supplies, Experiment 6 provides a deeper understanding of DC power supply fundamentals by observing the characteristics of half-wave rectification with a wall adapter transformer [1], [2]. Additionally, this experiment explores the idea of using a filter capacitor at the output to reduce ripple voltage and create a more stable DC output [2].

II. Background/Theory

As briefly discussed in the introduction, rectification is the operation of converting an alternating AC voltage into a signal that has only one polarity [1]. Diodes are especially useful in rectification due to the nonlinear characteristics; current flows for a particular polarity and the diode is off for another [1]. Instead of a DC voltage being analyzed, the source voltage will be a sinusoidal function denoted by

$$v(t) = V_m \sin(\omega t)\,, \tag{1}$$

where V_m refers to the peak amplitude of the input voltage, and ω is the angular frequency in radians per second ($\omega = 2\pi f$, where f is frequency in Hz) [2], [3]. Experiment 6 looks at both circumstances: half-wave rectification with and without a filter capacitor [2]. It is important to

understand with half-wave rectification that it is less efficient than full-wave rectification. Essentially, the negative portion of the sinusoidal input is cut off when the diode is not conducting. This produces a waveform displayed below in Fig. 1a, where the blue wave is the output voltage and the red wave refers to the input (line) voltage.

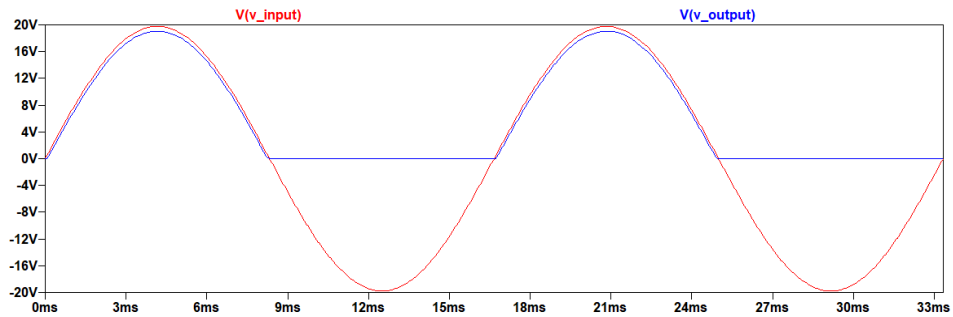

Fig. 1a. Waveform of half-wave rectification output versus input voltage.

A sinusoidal wave produces an average amplitude of zero over a period; therefore, the root mean square voltage (V_{rms}) will be used to characterize the AC input [2]. The root mean square value for a sinusoidal function can be related to the peak value by

$$V_m = \sqrt{2}V_{rms}. \tag{2}$$

The unidirectional output voltage produces a DC component of interest [2]. Since the waveform is periodic, the DC component can be derived as the average value of the function waveform [2]. This voltage is a function of angular frequency and time. Therefore, $v_o(\omega t)$ will produce a DC component calculated using the following derivation:

$$V_{DC} = \frac{1}{T}\int_0^T v_o(t)dt = \frac{1}{2\pi}\int_0^{2\pi} v_o(\omega t)d(\omega t)\,, \tag{3}$$

where T refers to the period (in seconds) for a full waveform cycle [2]. As derived in the Pre-lab section of the Appendix, this equation can be simplified to

$$V_{DC} = \frac{V_m - V_\gamma}{\pi}, \tag{4}$$

where the nominal voltage drop (V_γ) of 0.7 V from the diode is considered [2]. This parameter along with the input frequency (60 Hz) determines the output characteristics and the associated harmonics [2]. Harmonics are not a major topic to the general study performed in this experiment [2]. As illustrated in Fig. 1a, the diode does not conduct during a negative input voltage. This can be written as the following piecewise function:

$$v_o(\omega t) = \begin{cases} V_m \sin(\omega t) - V_\gamma, & \text{when the diode is conducting} \\ 0, & \text{when the diode is not conducting.} \end{cases} \tag{5}$$

These harmonics alter the DC signal with large amplitudes [2]. As briefly mentioned above, filtering the output's harmonics allows for a smoother DC signal. This is achieved by connecting a capacitor parallel to the load [1], [2]. Due to the capacitor's impedance, $Z_C = \frac{1}{j\omega C}$, decreasing as the frequency increases, the higher-frequency harmonics will 'pass' through the capacitor, filtering the output component as a smoother DC voltage [2]. Figures 1b and 1c below illustrate the differences in a half-wave rectifier circuit without and with a filter capacitor, respectively. In both figures, the source represents the secondary transformer step down voltage, and the resistor represents the load.

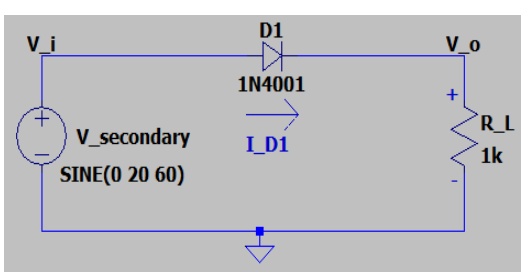

Fig. 1b. Schematic for a half-wave rectifier without a filter capacitor.

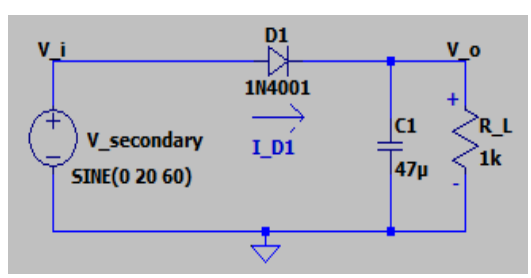

Fig. 1c. Schematic for a half-wave rectifier with a filter capacitor.

Even with a filter capacitor, the output waveform is still not a perfectly smooth DC signal [2]. As the diode conducts, the capacitor charges to a peak value of

$$V_{C,max} = V_m - V_\gamma. \tag{6}$$

When the diode is reverse-biased, the capacitor discharges exponentially through the parallel load resistor shown in Fig. 1c. This cycle repeats periodically, and the difference between the maximum and minimum output voltage is defined as the ripple voltage (V_R) [2]. For a half-wave rectification circuit without a filter, V_R is given by

$$V_R = V_m - V_\gamma. \tag{7}$$

With a filter capacitor, the ripple voltage is approximated by

$$V_R \approx \frac{(V_m - V_\gamma)}{f R_L C}, \tag{8}$$

where C is the capacitance value and R_L is the value of the load resistance [2]. The DC component of this output voltage (for a filtered circuit) can be expressed as:

$$V_{DC} = (V_m - V_\gamma) - \frac{V_R}{2}. \tag{9}$$

The average load current and peak diode current with a filter are, respectively:

$$I_{L,av} = \frac{V_{DC}}{R_L}, \tag{10}$$

$$I_{DP} = \frac{V_m - V_\gamma - V_R}{R_L} + \frac{2\pi(V_m - V_\gamma)}{R_L}\sqrt{\frac{2V_m}{V_R}}. \tag{11}$$

Equation (11) is specific for the half-wave rectifier circuit with a filter capacitor [2]. In contrast, the peak diode current equation without a filter is

$$I_{DP} = \frac{(V_m - V_\gamma)}{R_L}. \tag{12}$$

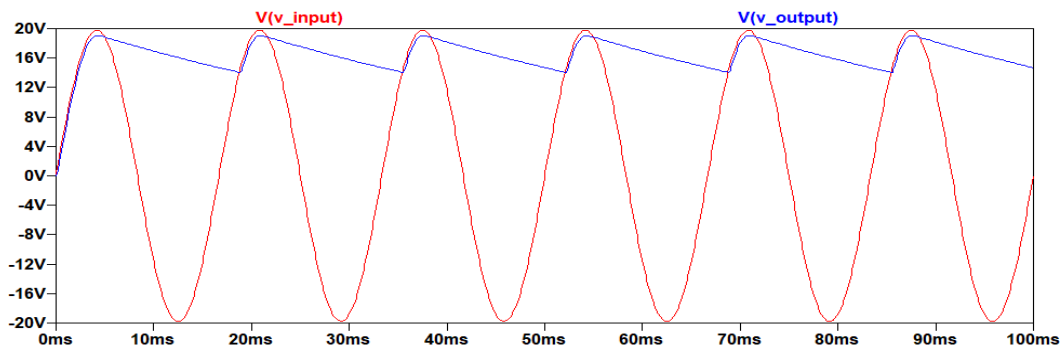

Fig. 1d. Waveform of half-wave rectification output versus input voltage with filter capacitor.

The resulting output waveform with a filter capacitor (Fig. 1d) can be compared to the unfiltered case (Fig. 1a). Capacitance values are directly correlated with the ripple; the larger the capacitance, the smaller the ripple, resulting in a more linear output [2]. However, larger capacitance may exceed the diode's peak current rating, which could result in damaging the device [2]. These parameters denote operational limits for half-wave rectification in this experiment. For more information regarding the theory of half-wave rectification, refer to [1] and [2] for further details.

III. Procedure

Circuit Schematics

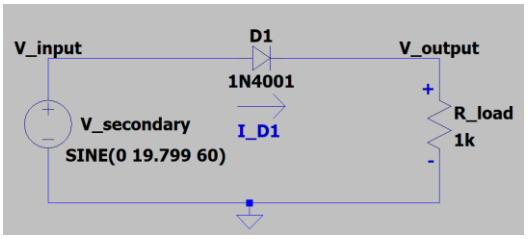

Fig. 2a. Schematic for half-wave rectifier without filter capacitor in Experiment 6.

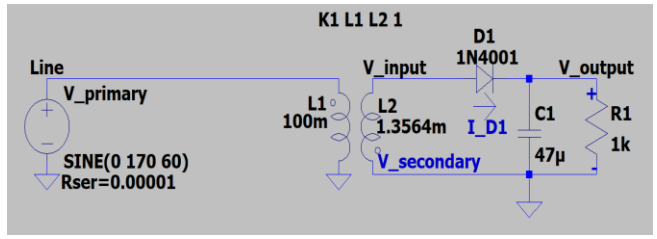

Fig. 2b. Schematic for a half-wave rectifier with filter capacitor (and transformer) in Experiment 6.

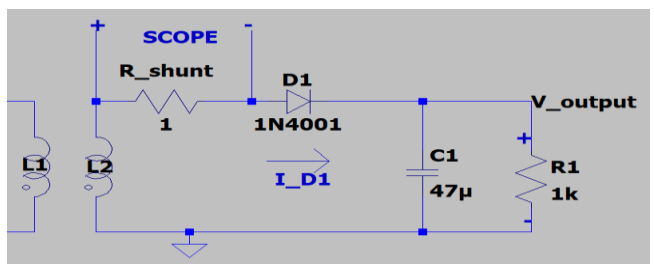

Fig. 2c. Schematic for a half-wave rectifier demonstrating how to apply a shunt resistor.

Pre-lab Data

Before attempting the physical experiment, use LTspice simulation software as well as the background theory from above to compute values of interest. Use different load resistors with values 195 Ω, 1.0 kΩ, and 4.7 kΩ. The capacitor will be 47 µF, and the diode model will be 1N4001 with a nominal voltage drop (V_γ) of 0.7 V. Using (2), compute a peak amplitude voltage (V_m) that corresponds to the wall adapter transformer of interest; in this case, the V_{rms} is 14 V at a frequency (f) of 60 Hz [2]. Find the DC components (V_{DC}) of both circuit figures (Fig. 2a and 2b) as well as the ripple voltages (V_R) using (4), (7), (8), and (9). Consult the 1N4001 datasheet [4] to determine the important diode component limits mentioned in the background theory and design. These include but are not limited to the following: peak repetitive reverse voltage,

average rectified forward current, and non-repetitive peak surge current [4]. For the half-wave rectifier circuit with the filter capacitor, take note of the peak voltage across the capacitor using (6), denoted by $V_{C,max}$. Once the data of interest has been computed, verify each calculation with LTspice software and take note of any discrepancies [2]. Ensure the diode limits are not exceeded. Refer to [2] to find the LTspice software directive that will successfully simulate the 1N4001 diode model used in this experiment. Additionally, the Pre-lab section of the Appendix will share how to simulate the transformer needed for Fig. 2b.

Preparation

Prior to beginning the experiment, ensure proper calibration in all the equipment used. The required instrumentation includes an oscilloscope, digital multimeter, solderless breadboard, wall adapter transformer, and generic test leads and probes. Specific manufacturer models and serial numbers are detailed in the Equipment section of the Appendix. A 47 μF capacitor as well as various resistors from the pre-lab data (195 Ω, 1.0 kΩ, and 4.7 kΩ) will be used. The 47 μF capacitor will be electrolytic; therefore, ensure proper polarity in both the diode (1N4001 model) and capacitor (47 μF). Electrolytic capacitors can suddenly explode if not handled appropriately. Reversing the polarity across the diode will cause the polarity across the capacitor to also be reversed [2]. It is vital to note to use two resistors with values of 390 Ω in parallel to achieve 195 Ω so the power dissipation limit rating of the resistors is not exceeded during the actual lab experiment [2].

A digital multimeter will be used to measure the average DC components of each circuit. Similarly, the oscilloscope will be used to measure peak voltage points due to the waveforms not being perfectly sinusoidal [2]. It is important to have both DC and AC components together

when measuring with the oscilloscope. Therefore, ensure the setting of the oscilloscope is DC coupled with all measurements [2]. For safety reasons, it is recommended to measure one waveform at a time when using the oscilloscope and one DC value at a time to ensure circuit integrity and prevent loose connections, which may lead to short circuits or component damage.

Experimental Step 1

Assemble the circuit provided by Fig. 2a above using a solderless breadboard. The voltage source will be the wall adapter transformer, the diode model will be 1N4001, and the load resistor (R_L) will be 1.0 kΩ. Figure 2f in the Procedure section of the Appendix shares the wall adapter transformer used in this experiment, however, the experiment is not completely biased on this model. The AC component of the transformer states 12 V (V_{rms}), however, this the output will be around 14 V for the RMS measurement. The rated value specifies under heavy load while this experiment will not deliver such a heavy load to the transformer [3]. Therefore, if the output matches the desired output (V_{rms}) voltage, any manufactured wall adapter transformer will work for the experiment. Before powering the circuit, cross-check the diode polarity [2]. Use the digital multimeter and a pair of test leads to take note of the root mean square voltage (V_{rms}) to verify the transformer's voltage (since the input is sinusoidal). Now, verify with the digital multimeter the DC voltage component of the load (V_{DC}) [2]. The oscilloscope will be used to verify the AC components (V_m) [2]. Take a screenshot of the oscilloscope and store the waveform on a USB drive. If this is not possible, use another method to document the display of the oscilloscope. Use the digital multimeter to determine the DC current component ($I_{L,av}$) across the load [2].

Experimental Step 2

Construct the circuit provided by Fig. 2b. Again, the voltage source will be the wall adapter transformer, the diode model will be the 1N4001, and the load resistor will be 1.0 kΩ. The capacitor will have a value of 47 μF. Once again, before powering the circuit, double check the polarity of the diode as well as the filter capacitor. Use the oscilloscope to plot the input waveform voltage and the load output voltage [2]. Again, use a USB drive to save this data for future analysis if possible. If that is not possible, use a different method to document the oscilloscope display. Write down the peak amplitude (V_m) AC component of each waveform like Experimental Step 1 [2]. Use the digital multimeter to measure the DC component of the load voltage (V_{DC}). Document the multimeter display for future verification, if needed. After all these data points have been taken, use a shunt resistor (1 Ω is recommended) to plot the waveform of the diode's current to measure the peak current across the diode (I_{DP}) [2]. Figure 2c above shows a LTspice simulation on how to apply a shunt resistor. A 1 Ω shunt resistor (R_{shunt}) will allow a near one-to-one ratio of current due to Ohm's Law ($V = IR$):

$$V_{peak} = I_{DP} R_{shunt} \tag{13}$$

$$\frac{V_{peak}}{R_{shunt}} = I_{DP} \quad \rightarrow \quad \frac{V_{peak}}{(1)} = I_{DP}$$

$$V_{peak} = I_{DP},$$

where V_{peak} is the peak voltage measured across the shunt resistor (R_{shunt}) and I_{DP} is the peak diode current. The waveform on the oscilloscope will display the diode's current [2], [3]. Take a screenshot of the waveform using the oscilloscope or use another method to document the display; then, take note of the diode's peak current (I_{DP}).

Experimental Step 3

Repeat the procedures in Experimental Step 2, substituting the load resistance with 4.7 kΩ [2].

Experimental Step 4

Repeat the procedures in Experimental Step 2, using a load resistance of 195 Ω. Connect two 390 Ω resistors in parallel to achieve a load resistance of 195 Ω so the power dissipation limit rating of the resistors used is not exceeded [2].

Disassembly

After completing the experiment, disconnect the transformer from the wall outlet first before touching the test leads or probes. Ensure the 47 μF is discharged before removing components from the breadboard. Disassemble the circuit and clean up the workspace by storing all leads and miscellaneous equipment used in appropriate storage locations. Finally, ensure all equipment is off before exiting the laboratory.

IV. Results/Analysis

The experimental results were evaluated by comparing the measured data against the theoretical values derived in the Pre-lab section of the Appendix. To quantify the accuracy, the percentage error for each parameter was calculated using the following relationship:

$$\%\,Error = \left|\frac{Measured - Theoretical}{Theoretical}\right| \times 100. \tag{14}$$

Table 1: Unfiltered Half-Wave Rectifier ($R_L = 1.0\ \text{k}\Omega$)

Parameter	*Theoretical*	*Simulated*	*Measured*	*% Error*
$V_{rms,in}$ (V)	14.000	14.000	13.902	0.700
$V_{m,in}$ (V)	19.799	19.773	20.600	4.046
$V_{m,out}$ (V)	19.099	19.099	19.400	1.576
$V_{DC,out}$ (V)	6.079	5.962	5.963	1.908
$V_{R,out}$ (V) [a]	19.099	19.053	19.400	1.576
$I_{L,av}$ (mA)	6.079	5.963	6.075	0.066

[a] Measured ripple ($V_{R,out}$) was derived from the measured peak voltage ($V_{m,out}$) with (7).

Table 1 summarizes and compares the primary results of Experimental Step 1. As shown, the data demonstrates high correlation with the theoretical computations, exhibiting negligible errors. This validates that (1)−(5) and (7) accurately model both simulation and physical circuit behavior. An important note here is that the average DC component is very low at 5.963 V. This is due to the no filter capacitor component in this circuit [2]. In the next step, the voltage will increase with the filter capacitor smoothing the voltage ripple [1]. The corresponding output and input waveforms are displayed in Fig. 3a as well as Fig. 3b, respectively. Figures 3c and 3d below display the LTspice software waveforms for a direct comparison to the oscilloscope [2]. For additional data and figures related to this step, refer to the Analysis section of the Appendix.

Fig. 3a. Input voltage waveform from Step 1.

Fig. 3b. Output voltage waveform from Step 1.

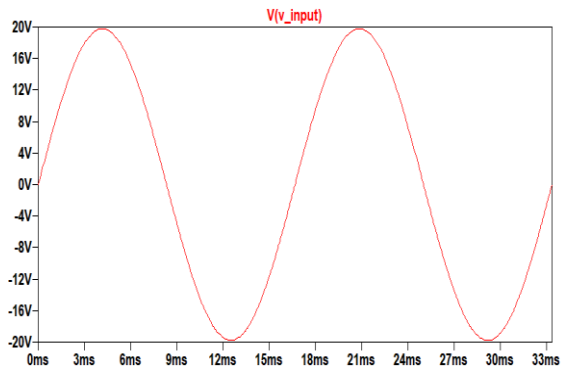

Fig. 3c. Input voltage waveform in LTspice software from Step 1.

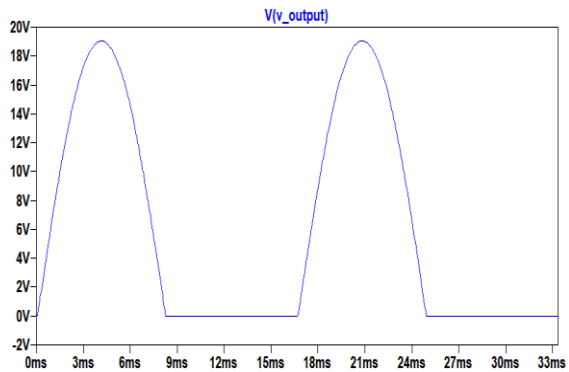

Fig. 3d. Output voltage waveform in LTspice software from Step 1.

Table 2: Filtered Half-Wave Rectifier ($R_L = 1.0$ kΩ)

Parameter	*Theoretical*	*Simulated*	*Measured*	*% Error*
$V_{m,in}$ (V)	19.799	19.799	20.400	3.036
$V_{m,out}$ (V)	19.099	19.060	19.400	1.576
$V_{DC,out}$ (V)	15.713	16.619	16.644	5.925
$V_{R,out}$ (V) [a]	6.773	5.047	6.879	1.565
I_{DP} (mA)	302.479	231.921	360.000	19.017

[a] Measured ripple ($V_{R,out}$) was derived from the measured peak voltage ($V_{m,out}$) with (8).

Table 2 compares the theoretical, measured, and simulated results for Experimental Step 2. There is noticeable increase in the error, particularly by the peak diode current (I_{DP}) and the DC component (V_{DC}). These discrepancies are likely attributed to the non-ideal characteristics of the transformer under load and the resistances of the 1N4001 diode, which are not fully accounted for in the linear approximated (9) and (11). Additionally, the capacitor will significantly influence the observed peak current, average output voltages, and ripple voltages. As shown above, the ripple voltage V_R has decreased while the DC component has increased. The capacitor filters the AC input accordingly allowing for a smoother DC signal [2]. Figures 4a, 4b, and 4c display the input waveform voltage, output waveform voltage, and diode current waveform respectively. Figures 4d, 4e, and 4f share the LTspice simulation software waveforms to compare directly to the oscilloscope measurements. Each waveform appears to directly correlate as intended. The output voltage waveform is zoomed in and slightly sharper, representing the smoothing of the filter capacitor. For additional figures, refer to the Analysis section of the Appendix.

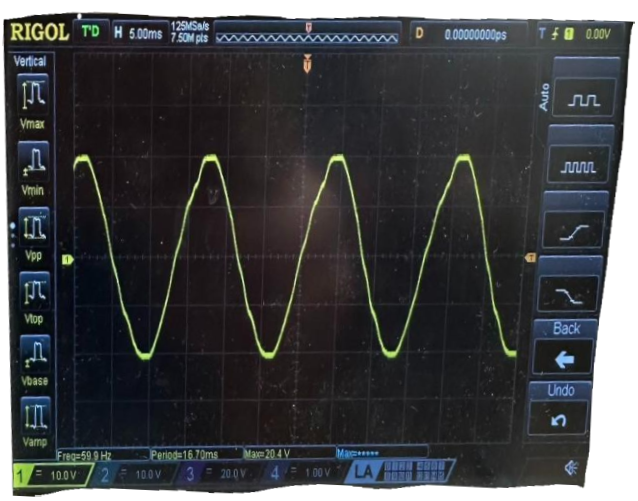

Fig. 4a. Input voltage waveform from Step 2.

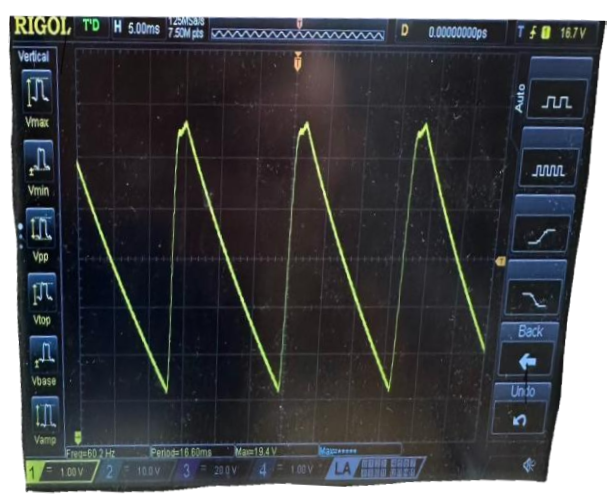

Fig. 4b. Output voltage waveform from Step 2.

Fig. 4c. Diode current waveform from Step 2. (Scale: 100 mV = 100 mA)

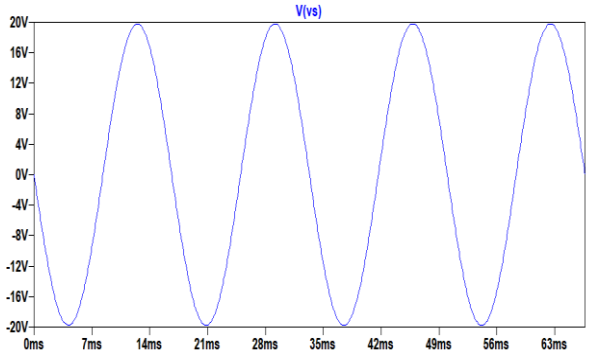

Fig. 4d. Input voltage waveform in LTspice software from Step 2.

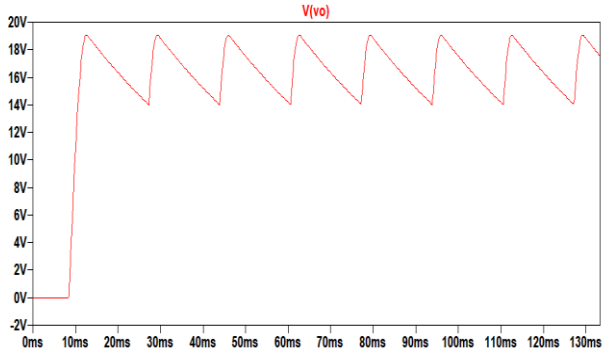

Fig. 4e. Output voltage waveform in LTspice software from Step 2.

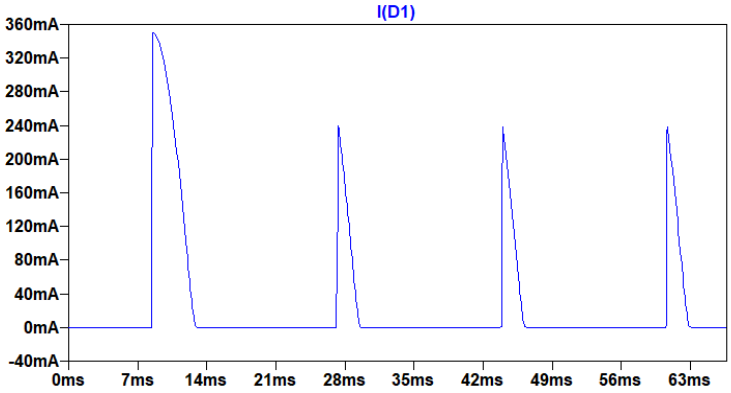

Fig. 4f. Diode current waveform in LTspice software from Step 2.

Table 3: Filtered Half-Wave Rectifier ($R_L = 4.7$ kΩ)

Parameter	*Theoretical*	*Simulated*	*Measured*	*% Error*
$V_{m,in}$ (V)	19.799	19.799	20.400	3.036
$V_{m,out}$ (V)	19.099	19.060	19.600	2.556
$V_{DC,out}$ (V)	18.364	18.428	18.716	1.917
$V_{R,out}$ (V) [a]	1.439	1.263	1.479	2.780
I_{DP} (mA)	137.484	100.076	160.000	16.377

[a] Measured ripple ($V_{R,out}$) was derived from the measured peak voltage ($V_{m,out}$) with (8).

Table 3 compares the theoretical, measured, and simulated results for Experimental Step 3. Like the previous step, there is a noticeable error with the diode peak current (I_{DP}) of 16.377 % while the other percentage errors are essentially negligible. This supports the conclusion that the transformer used appeared to be at a higher tolerance rate versus the expected tolerances from the theoretical calculations [2], [3]. Here, it is noticed as the resistance increases, the more DC voltage that will be a part of the output, hence $V_{DC} = 18.716$ V. The ripple voltage almost appears to decrease as the DC voltage increases, where $V_R = 1.479$ V. The linear (8)−(11) clearly work as the time constant ($\tau = R_L C$) remains a larger value [2]. The input voltage, output voltage, and diode current waveform are shown by Fig. 5a, 5b, and 5c, respectively. Each waveform appears to correspond accordingly to the simulated data from the Pre-lab section of the Appendix. Refer to the Analysis section of the Appendix for more figures of Experimental Step 3.

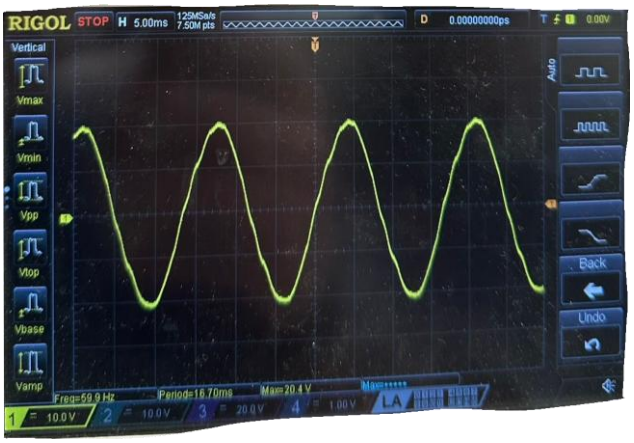

Fig. 5a. Input voltage waveform from Step 3.

Fig. 5b. Output voltage waveform from Step 3.

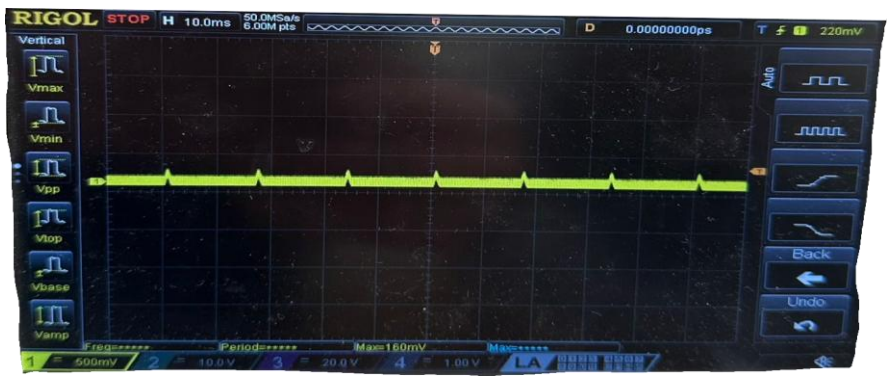

Fig. 5c. Diode current waveform from Step 3. (Scale: 100 mV = 100 mA)

Table 4: Filtered Half-Wave Rectifier ($R_L = 195\ \Omega$)

Parameter	*Theoretical*	*Simulated*	*Measured*	*% Error*
$V_{m,in}$ (V)	19.799	19.799	20.400	3.036
$V_{m,out}$ (V)	19.099	19.060	19.100	0.005
$V_{DC,out}$ (V)	1.732	11.128	11.169	544.861
$V_{R,out}$ (V) [a]	34.695	14.210	34.734	0.112
I_{DP} (mA)	576.342	356.067	440.000	23.656

[a] Measured ripple ($V_{R,out}$) was derived from the measured peak voltage ($V_{m,out}$) with (8).

Table 4 compares the theoretical, measured, and simulated results for Experimental Step 4. There is a massive discrepancy in the DC component (V_{DC}) and peak diode current (I_{DP}) with errors of 544.861 % and 23.656 %, respectively. This is where the linear approximated equations from the theory do not fully work as intended [2]. The discrepancy observed at the 195 Ω load is explained by the violation of the assumption of a higher time constant versus the period [2], [3]. The equation for the time constant and period can be denoted by

$$\tau = R_L C, \tag{15}$$

$$T = \frac{1}{f}, \tag{16}$$

where R_L is the load resistance, C is the capacitance value, and f is the frequency. Using (15), the time constant for this load is $\tau_{195} = 9.165$ ms while the time constant for the load in Experimental Step 3 (4.7 kΩ) is $\tau_{4700} = 220.900$ ms. Using (16), the period for each circuit was described to be $T = 16.667$ ms by the frequency (60 Hz). The time constant of 9.165 ms versus a period of 16.667 ms results in a non-linear approximation due to the characteristics of the diode in the half-wave rectifier circuit [2], [3]. This becomes an exponential approximation versus a linear one, resulting in large inaccuracies in the linear models (9) and (11) [3]. The ripple voltage (V_R) remains relatively accurate due to the diode being reverse-biased as the capacitor discharges [3]. Figures 6a, 6b, and 6c share the input voltage, output voltage, and diode current waveforms, respectively. The diode current waveform compares to the simulation data from the Pre-lab section in the Appendix. The measured peak diode current (440.0 mA) appears to run slightly high, which aligns with the previous tables with slightly higher characteristics [3]. Refer to the Analysis section of the Appendix for more figures and data regarding Experimental Step 4.

Fig. 6a. Input voltage waveform from Step 4.

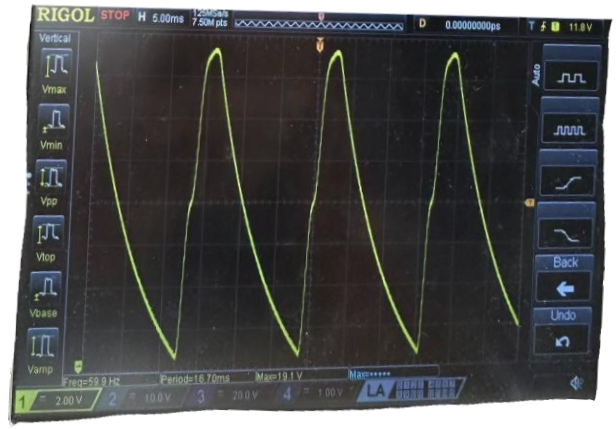

Fig. 6b. Output voltage waveform from Step 4.

Fig. 6c. Diode current waveform from Step 4. (Scale: 100 mV = 100 mA)

V. Conclusion

The laboratory objectives were successfully met through observing the response of a half-wave rectification circuit and comparing it to the computer-based simulation [2]. Experimental accuracy was achieved with minor discrepancies attributed to non-ideal component characteristics and higher tolerances compared to theoretical computations. This experiment demonstrated proper measurement techniques for basic DC power supplies by capturing key parameters used in voltage rectification [2]. Most notably, a discrepancy with the 195 Ω load resistance in the power supply circuit. While the simulation differed from the manual

calculations due to the linear approximations used by the lab documentation, the simulation still was validated by the physical measurements of the components [2], [3]. This justifies that while linear approximated circuit equations are useful for higher load scenarios, computer-based simulations provide a more accurate depiction of circuit operation when dealing with a non-linear diode component under various load resistances [2], [3].

VI. References

[1] Donald A. Neamen, *Microelectronics: Circuit Analysis and Designs*, 4th ed. New York, NY, USA: Neamen, 2010.

[2] Circuit and Electronics Lab, *Lab Manual* 6*: Half-Wave Rectifier*, Tennessee Technological University, 2026.

[3] C. Johnson. ECE 3050. Class Lecture, Topic: "Half Wave Rectification." Department of Electrical and Computer Engineering, Tennessee Technological University, Cookeville, TN, Mar. 10, 2026

[4] Onsemi, "Axial-Lead Glass Passivated Standard Recovery Rectifiers," 1N4001, 1N4002, 1N4003, 1N4004, 1N4005, 1N4006, 1N4007 datasheet, Feb. 2005 [Revised June 18, 2024].

VII. Appendix

Equipment

Manufacturer	*Model*	*Serial Number*	*Device*
Siglent	SDG2042X	SDG2XCAX4R0509	Function Generator
Keysight	34461A	MY59002200	Digital Multimeter
Rigol	MSO1104	DS1ZC171600509	Oscilloscope
Agilent	HP9100	–	Oscilloscope Probe
Multi-Link Inc.	AA-121A	–	Wall Adapter Transformer
–	Generic	–	Test Multimeter Leads
–	Generic	–	Solderless Breadboard
–	Generic	–	Resistors (2 x 390 Ω, 1.0 kΩ, 4.7 kΩ)
–	Generic	–	Capacitor (47 μF)

Procedure

Experimental Setup – Additional Data

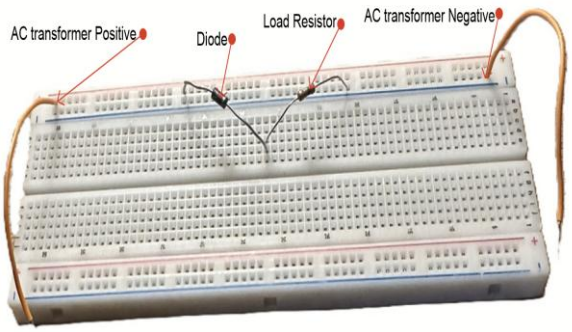

Fig. 2d. Diagram for half-wave rectifier without capacitor.

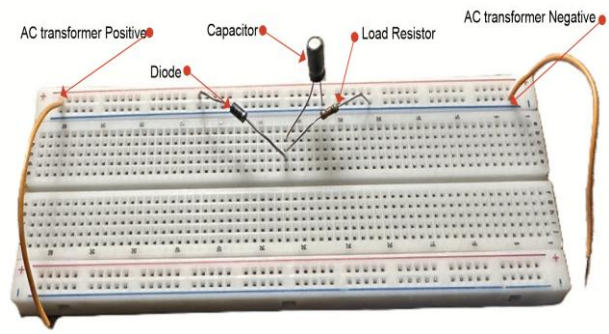

Fig. 2f. Diagram for half-wave rectifier with capacitor.

Fig. 2e. Picture of the wall adapter transformer used.

Analysis

Experimental Step 1 – Additional Data

Fig. 3e. Multimeter showing transformer's V_{rms}.

Fig. 3f. Multimeter displaying V_{DC}.

Fig. 3g. Multimeter displaying $I_{L,av}$.

Fig. 3h. Waveform of output versus input voltage (Step 1).

Experimental Step 2 – Additional Data

Fig. 4g. Multimeter displaying V_{DC}.

Fig. 4h. Waveform of diode voltage.

Experimental Step 3 – Additional Data

Fig. 5d. Multimeter displaying V_{DC}.

Fig. 5e. Waveform of diode voltage.

Experimental Step 4 – Additional Data

Fig. 6d. Multimeter displaying V_{DC}.

Fig. 6e. Waveform of diode voltage.